PHYSICAL CHEMISTRY

GUIDED REVISION PLAN_SCORE ADVANCED [SOLUTION]

Topic : ELECTROCHEMISTRY (Question Bank # 06)

SINGLE CORRECT

1. (C)

Sol. 100 g sample contazin 18.7 gm sodium (Na)

So, weight of Hg in 100 gm

Sample = 100 – 18.7 = 81.3 gm Hg

81.3 gm Hg $\longrightarrow$ 18.7 gm Na

$$10 \text{ gm Hg} \longrightarrow \frac{18.7}{81.3}\times 10$$

$$= 2.3 \text{ gm Na}$$

$$2.3 \text{ gm Na} \times \frac{1 \text{ mol Na}}{23 \text{ gm Na}} = 0.1 \text{ mol Na}$$

$$Na^+ + e^- \longrightarrow Na(s)$$

0.1 mol 0.1 mol

Since current efficiency = 50%

So charge required = 0.1 × 2 = 0.2 F

2. (B)

Sol. $\wedge_m^\infty (BaSO_4) = \wedge_m^\infty (BaCl_2) + \wedge_m^\infty (H_2SO_4) - \wedge_m^\infty (2HCl)$

$= (x_1 + x_2 - 2x_3)$ s cm^2 mol^{-1}

Specific conductance ($\kappa J = y\, s \times cm^{-1}$)

$$\wedge_m^\infty = \frac{\kappa \times 1000}{s}$$

$$s = \frac{\kappa \times 1000}{\wedge_m^\infty} = \frac{y \times 1000}{(x_1 + x_2 - 2x_3)}$$

$$BaSO_4 \rightleftharpoons Ba^{2+}_{(aq)} + SO^{2-}_{4\ (aq)}$$

$$k_{sp} = s^2 = \frac{10^6 y^2}{(x_1 + x_2 - 2x_3)^2}$$

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3. (A)

Sol. $\kappa = 9 \times 10^{-6}\ s\ m^{-1}$

$1\ s\ m^{-1} = 10^{-2}\ s\ cm^{-1}$

$\kappa = 9 \times 10^{-8}\ s\ cm^{-1}$

$\wedge_m^\infty = \wedge_{eq}^\infty \times n - factor$

$= 1.5 \times 10^{-4} \times 10^4 \times 3\ scm^2mol^{-1}$

$= 4.5\ s\ cm^2\ mol^{-1}$

$$\wedge_m^\infty = \frac{\kappa \times 1000}{s}$$

$$s = \frac{\kappa \times 1000}{\wedge_m^\infty} = \frac{9 \times 10^{-8} \times 1000}{4.5}$$

$= 2 \times 10^{-5}\ M$

$$Ag_3PO_{4(s)} \rightleftharpoons 3Ag^+_{(aq)} + PO^{3-}_{4(aq)}$$

$\quad s \qquad\qquad 3s \qquad\quad s$

$k_{sp} = 27\ s^4$

$= 2.7 \times (2 \times 10^{-5})^4$

$= 4.32 \times 10^{-18}$

4. (C)

Sol. Reaction at cathode :-

$$Cr_2O_7^{2-} + 14H^+ + 6e^- \longrightarrow 2Cr^{3+} + 7H_2O$$

$1\ M \qquad\qquad\qquad -$

$-1/2\ M \qquad\qquad 1\ M$

$\frac{1}{2}M \qquad\qquad\qquad 1\ M$

Nernst equation :-

$$E_{cell} = E^o_{cell} - \frac{0.06}{n}\log_{10} Q$$

$$= 1.33 - \frac{0.06}{6}\log\frac{\left[Cr^{3+}\right]^2}{\left[Cr_2O_7^{2-}\right]\left[H^+\right]^{14}}$$

$$= 1.33 - 0.01\log\left[\frac{1^2}{0.5\times(0.1)^{14}}\right]$$

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$= 1.33 - (0.01)(14.3)$

$= 1.33 - 0.143$

$= 1.187$

5. (B)

Sol. $\kappa = 5.5 \times 10^{-7}\ s\ cm^{-1}$

$$\Lambda_m^\infty (H_2O) = \lambda_m^\infty (H^+) + \lambda_m^\infty (OH^-)$$

$= 350 + 200$

$= 550\ s\ cm^2\ mol^{-1}$

$$\Lambda_m^\infty = \frac{\kappa \times 1000}{M}$$

$$M = \frac{\kappa \times 1000}{550} = \frac{5.5 \times 10^{-7} \times 1000}{550}$$

$= 10^{-6}\ M$

$k_w = M^2 = 10^{-12}$

$P^H = P^{OH} = 6$

6. (B)

Sol. $\text{Efficiency} = \frac{|W|}{q_{abs}} \times 100$

For electrochemical cell $\rightarrow \Delta G = W$

$$\Delta^\circ G = \Delta_f G^\circ (CO_2) + 2 \times \Delta_f G^\circ (H_2O) - \Delta_f G^\circ (CH_3OH) - \frac{3}{2} \times \Delta_f G^\circ (O_2)$$

$= -394.4 + 2(-237.2) + 166.2$

$= -702.6\ kJ\ mol^{-1}$

$\Delta H = q = -726\ kJ\ mol^{-1}$

$$\%\eta = \frac{\Delta G}{\Delta H} \times 100 = \frac{702.6}{726} \times 100$$

$= 97\%$

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MULTIPLE CORRECT

7. (A,B,D)

Sol. Fact Based

8. (A,C,D)

Sol. **Anode Reaction :-**

$$Hg_2^{2+} \longrightarrow 2Hg^{2+} + 2e^-$$

Cathode Reaction :-

$$NO_3^- + 2e^- + 3H^+ \rightarrow HNO_2 + H_2O$$

Overall Reaction :-

$$Hg_2^{2+} + 3H^+ + NO_3^- \rightarrow 2Hg^{2+} + HNO_2 + H_2O$$

$$E_{cell} = E^o_{cell} - \frac{0.06}{2}\log\frac{\left[Hg^{2+}\right]^2\left[HNO_2\right]}{\left[H^+\right]^3\left[Hg_2^{2+}\right]}$$

$$E_{cell} = E^o_{cell} - 0.03\ \log\frac{1}{[H^+]^3}$$

$$O = 0.02 - 0.03\times 3\ \log\frac{1}{\left[H^+\right]}$$

$$pH = \frac{2}{9}$$

9. (B,C)

Sol. Species having higher reduction potential will get reduced while species having lower reduction potential will get oxidised.

10. (A,B,C,D)

Sol. Fact based.

PARAGRAPH

11. (C)

Sol. (C) Since $KMnO_4$ also oxidise Cl^-.

12. (A)

Sol. eq. (FeC_2O_4) = eq. $(K_2Cr_2O_7)$

millimol $\times 3 = 0.1 \times 50 \times 6$

millimol $(FeC_2O_4) = 10$ millimol

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13. (B)

Sol. Equivalence of (Cl_2) + equivalence of (FeC_2O_4) = equivalence of $KMnO_4$

m. eq. (Cl_2) + m. eq. (FeC_2O_4) = m. eq. $(KMnO_4)$

$3.5 \times 2 + 3 \times x = 50 \times 0.1 \times 5$

$x = 6$ millimol

INTEGER

1. (0.19)

Sol. $Ag_2S \rightleftharpoons 2Ag^+ + S^{2-}$ $\quad \Delta G_1^0$

$2Ag^+ + 2e^- \rightleftharpoons 2Ag$ $\quad \Delta G_2^0$

$Ag_2S + 2e^- \rightleftharpoons 2Ag + S^{2-}$ $\quad \Delta G_3^0$

$\Delta G_3^0 = \Delta G_1^0 + \Delta G_2^0$

$(-2)\,f\,\varepsilon^o = -RT \ln k_{sp} + (-2)\,f\,(0.8)$

$$\varepsilon^o = \frac{2.303RT}{2F}\log k_{sp} + 0.8$$

$= 0.03 \log (10^{-49}) + 0.8$

$= -0.67$ V

$H_2S_{(aq)} \rightleftharpoons 2H^+_{(aq)} + S^{2-}_{(aq)}$

$$ka_1 \times ka_2 = \frac{[H^+]^2[S^{2-}]}{[H_2S]}$$

$$10^{-21} = \frac{(10^{-3})^2[S^{2-}]}{0.1}$$

$[S^{2-}] = 10^{-16}$ M

From Nernst equation :-

$$\varepsilon_{cell} = -0.67 - \frac{0.06}{2}\log[S^{2-}]$$

$= -0.67 - 0.03 \log (10^{-16})$

$= -0.67 + 0.48$

$= -0.19$ V

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2. (3.00)

Sol. Equivalent of Al formed

$$= \frac{Q}{F} = \text{moles} \times n_f$$

$$Al^{3+} + 3e^- \rightarrow Al\,(s)$$

$$\frac{0.1 \times 96.5 \times 3600 \times 1000}{96500 \times 10.8 \times 100} = \text{mol} \times 3$$

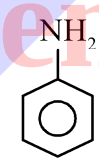
$$\text{mol} = \frac{0.1 \times 96.5 \times 3600 \times 1000}{96500 \times 10.8 \times 100}$$

$$\text{weight of Al} = \text{mol} \times \text{molar mass}$$

$$= \frac{0.1 \times 96.5 \times 3600 \times 1000}{96500 \times 10.8 \times 100} \times 27 = 3$$

3. (4.00)

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Sol. Graph (II) is for weak electrolyte CH_3COOH,  ,  , H_3PO_4

4. (8.00)

Sol. $X_2O_4^{2-} \longrightarrow$ New compound

$$\frac{Q}{F} = \text{mol} \times n - \text{factor}$$

$$\frac{965 \times 100}{96500} = 0.1 \times n_f$$

$$n_f = 10$$

Since process is oxidation, oxidation number of x in new compound will be greater than 3.

$$\text{so new oxidation state} = 3 + \left(\frac{10}{2}\right) = 8$$

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5. (0.54)

Sol. $Ag^+ + 2NH_3(aq) \rightleftharpoons Ag(NH_3)_2^+$

x 1M 0.01M

where $x \to 0$ since $k_f = 10^8$

$$k_f = \frac{0.01}{x \times 1^2} = 10^8$$

$x = 10^{-10}$ M

$[Ag^+]$ in cathode $= 10^{-10}$ M

$[Ag^+]$ in anode $= 0.1$ M

$$\varepsilon_{cell} = \frac{0.06}{1} \log \frac{[Ag^+]_{cathode}}{[Ag^+]_{Anode}}$$

$$= \frac{0.06}{1} \log \left(\frac{10^{-10}}{0.1} \right)$$

$= -0.54$ V

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SECTION-IV : MATRIX MATCH TYPE

1. (A-S, B-R, C-Q,T, D-P)

Sol. Fact based.

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